

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 13: Pythagoras Theorem

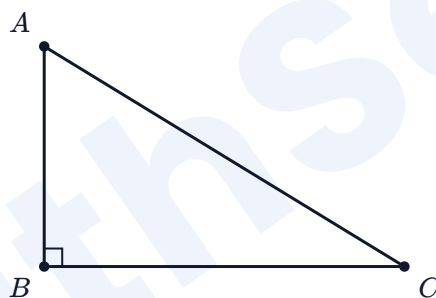
Topic-wise short notes with solved examples

Pythagoras theorem and its converse, Pythagorean triplets, and their use in finding lengths in triangles, rectangles, rhombuses and everyday situations.

## 1 The theorem

- In a **right-angled triangle**, the square on the **hypotenuse** equals the sum of the squares on the other two sides.
- If  $\angle B = 90^\circ$  in triangle ABC, then

$$AC^2 = AB^2 + BC^2$$



Angle B is  $90^\circ$ , so  $AC^2 = AB^2 + BC^2$

- The hypotenuse is the side opposite the right angle. It is the longest side.
- A proof by area:** draw BD perpendicular to AC. Triangles ABD and ACB have the same angles, which gives  $AB^2 = AD \times AC$ . In the same way  $BC^2 = DC \times AC$ . Adding:  $AB^2 + BC^2 = AC(AD + DC) = AC^2$ .

**Example 1.1** In triangle ABC,  $\angle B = 90^\circ$ ,  $AB = 9$  cm and  $BC = 12$  cm. Find AC.

**SOLUTION**

$$AC^2 = 81 + 144 = 225$$

**15 cm**

**Example 1.2** In triangle PQR,  $\angle Q = 90^\circ$ ,  $PR = 25$  cm and  $PQ = 7$  cm. Find QR.

**SOLUTION**

$$QR^2 = 625 - 49 = 576$$

**24 cm**

## 2 Converse

- If the square on one side of a triangle equals the sum of the squares on the other two sides, then the angle **opposite the first side** is a right angle.
- To test a triangle:** square the longest side and compare with the sum of the squares of the other two.

| Comparison (c is the longest side) | Triangle      |
|------------------------------------|---------------|
| $c^2 = a^2 + b^2$                  | Right-angled  |
| $c^2 < a^2 + b^2$                  | Acute-angled  |
| $c^2 > a^2 + b^2$                  | Obtuse-angled |

**Example 2.1** Is a triangle with sides 8 cm, 15 cm and 17 cm right-angled?

SOLUTION

$$17^2 = 289 \text{ and } 8^2 + 15^2 = 64 + 225 = 289$$

**Yes; the right angle is opposite the 17 cm side**

**Example 2.2** What kind of triangle has sides 5 cm, 6 cm and 9 cm?

SOLUTION

$$81 > 25 + 36 = 61$$

**Obtuse-angled**

## 3 Pythagorean triplets

- Three natural numbers  $a, b, c$  with  $a^2 + b^2 = c^2$  form a **Pythagorean triplet**.
- Common triplets: (3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25), (20, 21, 29), (9, 40, 41).
- Any multiple of a triplet is a triplet: (6, 8, 10), (9, 12, 15), (10, 24, 26).
- For any natural number  $m > 1$ :  $(2m, m^2 - 1, m^2 + 1)$  is a triplet.

**Example 3.1** Write a Pythagorean triplet one of whose members is 12.

SOLUTION

$$\text{Take } 2m = 12, \text{ so } m = 6: m^2 - 1 = 35 \text{ and } m^2 + 1 = 37.$$

$$(12, 35, 37)$$

## 4 Standard applications

- Diagonal of a rectangle:**  $d^2 = l^2 + b^2$ . **Diagonal of a square** of side  $a$ :  $a\sqrt{2}$ .
- Rhombus:** the diagonals bisect each other at right angles, so  $\text{side}^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$ , that is  $4 \times \text{side}^2 = d_1^2 + d_2^2$ .

- **Isosceles triangle:** the altitude to the base bisects the base.
- **Equilateral triangle** of side  $a$ : altitude  $= \frac{\sqrt{3}}{2}a$  and area  $= \frac{\sqrt{3}}{4}a^2$ .
- **Altitude on the hypotenuse:** if the legs are  $a, b$  and the hypotenuse is  $c$ , the altitude is  $\frac{ab}{c}$ .

**Example 4.1** A rectangle is 24 cm long and its diagonal is 26 cm. Find its breadth.

SOLUTION

$$b^2 = 676 - 576 = 100$$

**10 cm**

**Example 4.2** The diagonals of a rhombus are 16 cm and 30 cm. Find its perimeter.

SOLUTION

$$\text{Side}^2 = 8^2 + 15^2 = 289, \text{ so the side is } 17 \text{ cm.}$$

**68 cm**

**Example 4.3** Find the altitude of an equilateral triangle of side 8 cm.

SOLUTION

$$h^2 = 8^2 - 4^2 = 48$$

**$4\sqrt{3}$  cm, about 6.93 cm**

**Example 4.4** The legs of a right triangle are 6 cm and 8 cm. Find the altitude on the hypotenuse.

SOLUTION

$$\text{Hypotenuse} = 10 \text{ cm. Area} = \frac{1}{2} \times 6 \times 8 = \frac{1}{2} \times 10 \times h.$$

**4.8 cm**

## 5 Problems from daily life

- Draw the figure, mark the right angle and name the three sides before using the theorem.

**Example 5.1** A ladder 13 m long reaches a window 12 m above the ground. How far is the foot of the ladder from the wall?

SOLUTION

$$d^2 = 169 - 144 = 25$$

**5 m**

**Example 5.2** Two poles of heights 6 m and 11 m stand 12 m apart on level ground. Find the distance between their tops.

**SOLUTION**

The tops differ in height by 5 m and are 12 m apart horizontally:  $d^2 = 25 + 144$ .

**13 m**

**Example 5.3** A man goes 15 m due west and then 8 m due north. How far is he from the starting point?

**SOLUTION**

$$d^2 = 225 + 64 = 289$$

**17 m**

## 6 Results in triangles

- In triangle ABC with AD perpendicular to BC:  $AB^2 - BD^2 = AC^2 - CD^2$  (both equal  $AD^2$ ).
- If  $\angle B = 90^\circ$  and D is the mid-point of BC:  $AC^2 = AD^2 + 3CD^2$ .
- In an equilateral triangle with altitude AD:  $3AB^2 = 4AD^2$ .

**Example 6.1** In triangle ABC, AD is perpendicular to BC,  $AB = 13$  cm,  $BD = 5$  cm and  $DC = 9$  cm. Find AC.

**SOLUTION**

$$AD^2 = 169 - 25 = 144. \text{ Then } AC^2 = 144 + 81 = 225.$$

**15 cm**

**Example 6.2** In triangle ABC,  $\angle B = 90^\circ$  and D is the mid-point of BC. Prove that  $AC^2 = AD^2 + 3CD^2$ .

**SOLUTION**

$$AC^2 = AB^2 + BC^2 \text{ and } AD^2 = AB^2 + BD^2. \text{ Subtracting: } AC^2 - AD^2 = BC^2 - BD^2.$$

$$BC = 2CD \text{ and } BD = CD, \text{ so } BC^2 - BD^2 = 4CD^2 - CD^2.$$

$$AC^2 = AD^2 + 3CD^2$$

## 7 Common mistakes

- Adding when a shorter side is required: use  $\text{hypotenuse}^2$  minus the known side<sup>2</sup>.
- Taking the wrong side as the hypotenuse. It is always opposite the right angle.
- Forgetting to take the square root at the end.
- Using the full diagonals of a rhombus instead of the half-diagonals.
- Testing the converse with a side that is not the longest.
- Using the theorem in a triangle that is not right-angled.

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